A deep neural network just stacks these layers together, and we train it by applying gradient descent on the weight matrices, which are the parameters of the neural network. Here's a simple 2-layer neural network in Numpy.

Listing A.2. A simple neural network

```
def nn(x,w1,w2):
    l1 = x @ w1                 1
    l1 = np.maximum(0,l1)       2
    l2 = l1 @ w2
    l2 = np.maximum(0,l2)
    return l2

w1 = np.random.randn(784,200)   3
w2 = np.random.randn(200,10)
x = np.random.randn(784)        4
nn(x,w1,w2)

array([326.24915523,   0.        ,   0.        , 301.0265272 ,
       188.47784869,   0.        ,   0.        ,   0.        ,
         0.        ,   0.        ])
```

- ***1*** Matrix multiplication
- ***2*** Nonlinear activation function
- ***3*** Weight (parameter) matrix, initialized randomly
- ***4*** Random input vector

In the next section you'll learn how to use the PyTorch library to automatically compute gradients to easily train neural networks.